Algebra 1 Name___________________
Unit 9 Date____________________
Lesson 7 Homework Period___________________

1. For each expression, use the zero product property to determine the values of x for which the expression would equal 0.

Expression	$\boldsymbol{x(x-7)}$	$\boldsymbol{(x-5)(2x+8)}$	$\boldsymbol{(-5x-7)(3x+1)}$
$\boldsymbol{x}$**-values**			

Denise's work when solving the equation $6x^2 - 19x + 1 = 2x^2 + 6$ by factoring is shown. Use Denise's work to answer questions 2 and 3.

$6x^2 - 19x + 1 = 2x^2 + 6$
$6x^2 - 19x + 1 - (2x^2 + 6) = (2x^2 + 6) - (2x^2 + 6)$
$4x^2 - 19x - 5 = 0$
$(4x + 1)(x - 5) = 0$
$(4x + 1) = 0$ or $(x - 5) = 0$
$x = 4$ or $x = -5$

2. Circle the step where Denise made an error.

3. Correct Denise's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

4. Fill in the blanks to accurately finish solving the equation $3x^2 + 13x - 2 = 2x + 2$ by factoring.

$3x^2 + 13x - 2 = 2x + 2$
$3x^2 + 13x - 2 - (\_\_\_\_\_\_\_\_) = 2x + 2 - (\_\_\_\_\_\_\_\_)$
$3x^2 + 11x - 4 = \_\_\_\_$
$(3x - 1)(\_\_\_\_\_\_\_) = 0$
$(\_\_\_\_\_\_\_\_) = 0$ or $(\_\_\_\_\_\_\_\_) = 0$
$x = \_\_\_\_\_$ or $x = \_\_\_\_\_$

Solve each equation by factoring. Show your work.

5. $x^2 + 13x + 40 = 0$

6. $x^2 - 2x - 10 = 5$

7. $4x + 2 = 3x^2 - 7x + 12$

8. $9x^2 - 12x - 5 = 0$

9. $2x^2 - 4x - 70 = 0$

10. $-5 = 14x^2 + 17x$